

Praveen Korukonda

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Education

University of Chicago | *Chicago, IL, USA*

June 2026 – Dec 2027

Master of Science Computer Science

Indian Institute of Technology Roorkee | *Uttarakhand, India*

Oct 2021 – Sep 2025

Bachelor of Technology

Skills

Languages: C/C++, Java, Python, Golang, JavaScript, SQL

Technologies: Git, Docker, AWS, Spring Boot, REST APIs, CI/CD, React, MySQL, PostgreSQL, Linux

CS Fundamentals: System Design, Data Structures, Algorithms, Objected Oriented Programming

Experience

Software Engineering Intern | *Planto AI, India*

Sept 2025 – Dec 2025

- Built an incident management system using microservices architecture with Go and PostgreSQL database
- Designed 15+ REST APIs with unit tests across 4 services to manage end-to-end incident lifecycles
- Reduced Mean Time to Acknowledge (MTTA) by 27% through automated, real-time Slack alerts
- Set up background cron jobs for automated incident assignment and send frequent reminders
- Built an invoice generation service in Go to create dynamic PDF invoices, achieving 55 ms average latency
- Developed secure REST APIs and integrated PostgreSQL for storing and retrieving PDF invoices

SWE Intern | *CodingPro, India*

June 2025 – Aug 2025

- Built an end-to-end secure job portal application using Spring Boot, Thymeleaf and MySQL database
- Containerized the application using Docker and deployed it to AWS Elastic Beanstalk with RDS for storage
- Implemented role-based authentication with Spring Security to secure both candidate and recruiter profiles
- Developed e2e recruitment workflows, profile management, job application tracking, and resume uploads

Projects

byte-shell | *Association for Computing Machinery (ACM), IIT Roorkee*

praveenk.dev/byte-shell

- Built a Unix command line shell from scratch in C that used to execute built-in Linux commands
- Implemented core features like directory navigation and process handling using low-level unix system calls

mini-redis

praveenk.dev/mini-redis

- Developed an in-memory Redis-style server and a CLI client in Go that supports core Redis commands
- Implemented thread-safe storage for strings (with expiry), lists (with blocking operations), and sorted sets
- Supports multiple concurrent clients and loads an existing RDB file at startup for basic persistence

torrent-x

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- Created a BitTorrent client in Java that allows downloads using a single-file torrent or magnet links
- Implemented p2p protocol with tracker communication, piece verification, and metadata exchange

caching-proxy

praveenk.dev/caching

- Built a caching reverse proxy command-line tool in Go that reduces server hits and request latency
- Developed a thread-safe and concurrent disk-based cache storage system using SHA-256 hashing

git-lite

praveenk.dev/git-lite

- Created a Git client in Java that clones Github repositories using Git's Smart HTTP, which is simple and fast

uptime-monitor

praveenk.dev/uptime

- Built an automated uptime monitoring system in Go that continuously monitors the uptime of websites
- Integrated Slack webhook notifications via pub/sub pipeline to deliver automatic real-time downtime alerts

go-lb

praveenk.dev/go-lb

- Designed a load balancer in Go that supports concurrent request handling and active/passive health checks

Image Denoiser | *Vision and Language Group (VLG), IIT Roorkee*

[github/image-denoiser](https://github.com/image-denoiser)

- Designed an autoencoder-based image denoiser for low-light photos, reaching a PSNR of 28.256 dB
- Applied RIDNet to improve visibility and detail retention, raising PSNR to 30.579 dB on the same set

Stock Sentiment Analysis Using ML | *Finance Club, IIT Roorkee*

[github/stock-sentiment](https://github.com/stock-sentiment)

- Built an NLP-driven sentiment model to predict stock-price movements from financial news headlines
- Back-tested the trading strategy on historical data, yielding 17.12% returns and a 1.34 Sharpe ratio

Honors & Achievements

- An active member on Codeforces community (max. rating: 1641, expert), handle: "lite_27"
- Secured an All-India-Rank of 6026 in JEE Main 2021 amongst 1 Million candidates
- Secured 99.63 percentile in Math section of JEE Main 2021, placing me in the top 0.37% of the test takers
- Secured an All-India-Rank of 7514 in JEE Advanced 2021 amongst 250,000 candidates

Research Experience

Bachelor Thesis | Estimation of RUL of Bearings Using Machine Learning

Sep 2024 – May 2025

Supervisor: *Prof. S.P. Harsha, IIT Roorkee*

- Built a predictive-maintenance framework estimating Remaining Useful Life of high-speed bearings
- Collected vibration signals from an experimental test rig and denoised them via wavelet thresholding
- Extracted 34 multi-domain features and filtered them via S-weighted and Pearson correlation scoring
- Designed a self-attention-augmented BiLSTM to model temporal degradation patterns in vibration data
- Tuned hyperparameters via Bayesian optimisation, selecting hidden units, dropout, and learning rate
- Achieved R^2 of 0.825 on combined data and 0.781 on the held-out bearing under leave-one-out testing

Research Internship | Simulation of Human Induced Load Patterns Using ML

July 2024 – Aug 2024

Supervisor: *Prof. Anil Kumar, IIT Roorkee*

- Developed a GAN-based model generating human-induced load signals at 98% similarity to real data
- Processed 658+ experimental samples across walking, jumping, and bouncing loads at 1.5 to 3.5 Hz
- Built a Forecasting+LSTM model and bench marked ARIMA, Prophet for stochastic time-series synthesis
- Validated synthetic loads via Power Spectral Density matching and statistical equivalence testing

University Service

Technical Council Member | *Govind Bhawan Hostel, IIT Roorkee*

March 2024 – June 2024

- Collaborated with hostel administration to improve technical infrastructure and quality of internet services
- Organized technical workshops & hackathons to enhance technical skills among 600+ hostel students

Contingent Leader | *Student Technical Council, IIT Roorkee*

April 2024

- Led a contingent of 60 students, representing my hostel in the annual Inter Hostel Tech Fest 2024
- Led participation across 8 diverse technical Problem Statements & guided the teams to secure 4 Silver Medals